

Department of Physics and Astronomy, NC State University
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Stellar evolution, massive stars, core-collapse supernovae, supernova progenitors, astrophysical neutrinos

PhD in Physics, NC State University · Raleigh, NC expected 2027
 Thesis: *Effects of Stellar Evolution Uncertainties on Red Supergiant Progenitors and their Explosions*
 Committee: James P. Kneller (chair), Carla Fröhlich, Ian Roederer, John Blondin

MS in Physics, NC State University · Raleigh, NC awarded Dec 2023
 BA in Physics, NC State University · Raleigh, NC awarded May 2021
 BS in Science, Technology, and Society, NC State University · Raleigh, NC awarded May 2021

Meena and Pratul Ajmera Award [†]	2025
Sayers Fellowship [†]	2021
Mitchell Fellowship [†]	2021
Richard R. Patty Award for Leadership [†]	2021
Undergraduate Research Award*	2020

[†] Awarded by Department of Physics and Astronomy, NC State University

* Awarded by College of Sciences, NC State University

Total citations: 41 || h-index: 2 || [ORCID](#), [INSPIRE](#), [Google Scholar](#)

The Effects of Mass Loss and Convective Overshooting on the Pre-Collapse Structure, Composition, and Neutrino Emission of Red Supergiants.

M. A. Myers, C. B. Campbell, K. M. Patton, and J. P. Kneller.

Neutrino flavor mixing with moments.

M. Myers, T. Cooper, M. Warren, J. Kneller, G. McLaughlin, S. Richers, E. Grohs, & C. Fröhlich. (2022). Phys. Rev. D, 105, 123036. [10.1103/PhysRevD.105.123036](https://arxiv.org/abs/10.1103/PhysRevD.105.123036). Citations: 20

SNEWPY: A Data Pipeline from Supernova Simulations to Neutrino Signals.

A. L. Baxter, S. BenZvi, J. Castaneda, A. Coleiro, M. Colomer Molla, D. Dornic, S. Griswold, T. Goldhagen, A. Graf, A. Habig, R. Hill, S. Horiuchi, J. P. Kneller, M. Lamoureux, R. F. Lang, M. Lincetto, J. Migenda, M. Myers, E. O'Connor, A. Renshaw, K. Scholberg, A. Sheshukov, J. Tseng, C. Tunnell, N. Uberoi, & A. Worlikar.

(2021). Journal of Open Source Software, 6, 3772. [10.21105/joss.03772](https://doi.org/10.21105/joss.03772). Citations: 21

Selected presentations

- Effects of Mass Loss and Overshooting on the Pre-Supernova Neutrino Signal in Red Supergiants* Jul 2025
Invited seminar, Division of Elementary Particle-, Nuclear-, and Astrophysics, Institute of Theoretical Physics, University of Wroclaw · Wroclaw, Poland
- Effects of Mass Loss and Overshooting on the Pre-Supernova Neutrino Signal in Red Supergiants* Jul 2025
Invited talk, HELIUM25 · Dresden, Germany
- Effects of Mass Loss and Overshooting on the Pre-Supernova Neutrinos from Red Supergiants* Jun 2025
Contributed poster and flash talk, Nuclei in the Cosmos XVIII · Girona, Spain
- Effects of Mass Loss and Overshooting on the Pre-Supernova Neutrinos from Red Supergiants* May 2025
Contributed poster, Frontiers in Nuclear Astrophysics · Athens, OH
- The Effect of Mass Loss on Supernova Progenitors* Sep 2024
Seminar, SNEWS 2.0 collaboration · online
- Predicting the Next Galactic Core-Collapse Supernova* Mar 2023
Seminar, Department of Statistics, NC State University · Raleigh, NC

Mentorship

Undergraduate researchers

- Claire B. Campbell, Illinois State University May 2024–present
- Evolved red supergiants with the stellar evolution code MESA.
- Daniel Almonte, Stony Brook University May–August 2024
- Simulated earth matter effects on neutrino flavor oscillations.

Teaching experience

- Adjunct Faculty**, Wake Technical Community College Jan 2024–present
Tutoring in physics, mathematics, chemistry, computer programming, and writing.
- Teaching Assistant**, Department of Physics and Astronomy, NC State University
- PY 125: Astronomy Laboratory 4 semesters, 2022–2025
- Instruction in observing with the Skynet robotic telescope network, image analysis, and related software. Optical, UV, and radio observations of Solar System objects, galaxies, nebulae, globular clusters, and the Galactic center. Responsible for all grading, class policies, scheduling, and administering the curriculum.
- PY 209: Physics for Engineers and Scientists II Laboratory Fall 2021
- Average student rating: 4.58/5.00
 - Mini lectures, experiments, and exercises in electromagnetism to accompany the lecture course PY 208. Circuits and Python programming emphasized.

Service and outreach

- Secretary/Treasurer, Graduate Physics Student Association May 2023–May 2025
- Organized events and managed financial resources for the NC State University GPSA.
- Moore Square Middle School Dec 2023
- Taught children about sun spots and observed them together through a telescope.

APS-IDEA Group

Sep 2020–Aug 2022

- Participated in training and contributed to discussions about cultivating a positive and diverse work environment in the physics community.

Diversity, Equity, and Inclusion Committee

Sep 2020–Aug 2022

- Explored institutional resources to support DEI initiatives in NC State's physics department.

Secretary, Women in Physics

Aug 2020–Aug 2022

- Organized events and mentored undergraduate women in NC State's physics department.

References

Available upon request.